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Semantics versus Identity: A Divide-and-Conquer Approach towards Adjustable Medical Image De-Identification

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Abstract

Medical imaging has significantly advanced computer-aided diagnosis, yet its re-identification (ReID) risks raise critical privacy concerns, calling for de-identification (DeID) techniques. Unfortunately, existing DeID methods neither particularly preserve medical semantics, nor are flexibly adjustable towards different privacy levels. To address these issues, we propose a divide-and-conquer framework comprising two steps: (1) Identity-Blocking, which blocks varying proportions of identity-related regions, to achieve different privacy levels; and (2) Medical-Semantics-Compensation, which leverages pre-trained Medical Foundation Models (MFMs) to extract medical semantic features to compensate the blocked regions. Moreover, recognizing that features from MFMs may still contain residual identity information, we introduce a Minimum Description Length principle-based feature decoupling strategy, to effectively decouple and discard such identity components. Extensive evaluations against existing approaches across seven datasets and three downstream tasks, demonstrates our state-of-the-art performance.

1. Introduction

In the era of digital medicine, large-scale medical images, such as X-rays and fundus photographs [6], are routinely processed by AI-based diagnostic models [28, 87, 106, 119, 121] to aid clinical decision-making. However, the increasing availability of these images raises significant concerns regarding patient privacy [17, 56, 81, 89], calling for the research on medical image de-identification.

Although explicit personal details such as patient name can be easily removed from medical image headers [1, 73, 85] or burned-in texts [102, 123], re-identification (ReID) remains feasible for the intrinsic bio-identifiers, such as anatomical markers visible in chest X-rays [35, 76, 99].

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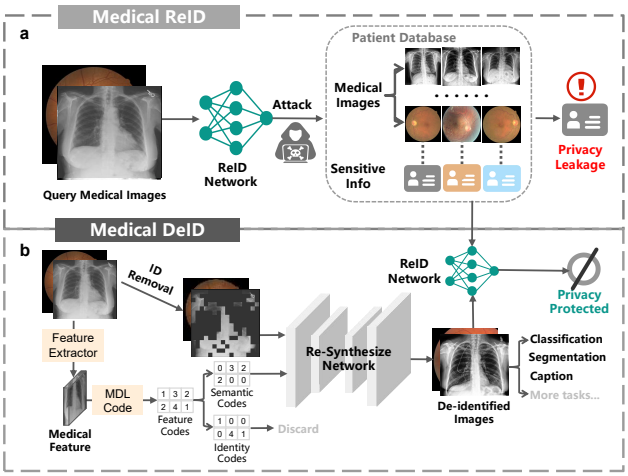

Figure 1. (a) Given the query medical image, the ReID model can retrieve sensitive patient information from a leaked database. (b) Our DeID framework, removing identity and then compensating medical semantics, ensures adjustable identity protection, while preserving downstream task utility. Besides, a Minimum Description Length (MDL) principle-based code space is introduced, to decouple and discard the identity information in medical features.

This enables sensitive information breaches [9, 52, 90], compromising patient privacy (see Figure 1(a)).

Several studies have attempted to defend against ReID attacks. For instance, some approaches [12, 30, 32, 44] focus on removing facial features to obfuscate identity. However, such methods cannot be applied to other body parts like the chest, where identity information is deeply interwoven with diagnostic semantics. Standard image filtering techniques, such as blurring [103], pixelation [41], and masking [108], indiscriminately degrade critical diagnostic details, thereby impairing downstream medical task performance. Moreover, under high privacy settings, the severe degradation of image quality further deteriorates task performance. Differential privacy methods [19, 27, 57, 112] mitigate identity information via noise injection, but this operation also perturbs diagnostic features. Identity adver-

sarial learning methods [55, 77] train generators by jointly maximizing identity discrepancy between the generated and original images, while minimizing distortion of medical information. Nevertheless, given the inherent entanglement between identity and diagnostic features, these methods fail to preserve diagnostic semantics at high privacy levels adequately. Recently, diagnostic annotation-conditioned generative models [15, 26, 43, 91, 109] have yielded promising results, yet they remain limited to task-specific semantics and cannot offer adjustable privacy levels. In summary, *no existing method preserves task-generalizable semantics, while supporting a wide range of adjustable privacy levels*.

To address these issues, we introduce a novel divide-and-conquer framework DCM-DeID, which decouples identity removal from semantic preservation, to achieve semantic-rich yet adjustable de-identification. Our approach includes three steps, i.e., *ID-Blocking*, which masks identity-related regions to achieve adjustable privacy levels; *Medical Semantics Extraction*, which leverages pre-trained medical foundation models (MFMs) [74, 117] to extract semantic-rich medical features; *Image Re-Synthesis*, which employs a diffusion model [42, 86] to synthesize de-identified images, given the above ID-masked image and the medical features. Moreover, considering that the features from MFMs may also contain some identity information, we introduce a novel minimum description length [33]-based feature decoupling strategy, which excludes identity-associated information from the vanilla MFM features in a minimum-codelength latent space. This effectively prevents the reintroduction of identity information during the image resynthesis step. Through the above designs, our approach achieves a better trade-off between DeID and diagnostic utilization than prior approaches. Our contributions are:

- We reveal that existing medical DeID methods fall short in preserving task-generalizable semantics, and do not adjust seamlessly across privacy levels. We build the first benchmark for this problem, by reproducing previous approaches fairly on seven datasets.
- We propose the DCM-DeID framework, which performs identity removal and medical semantics preservation in separate steps, enabling both adjustable privacy protection and medical task utility.
- We introduce a Minimum Description Length-based decoupling strategy, which decouples identity cues from medical features in a compact code space, further improving the privacy protection capability.
- Our framework demonstrates state-of-the-art performance. Extensive Analysis is also performed.

2. Related Works

Image Privacy Protection. Early methods applied low-level filters to obscure image details, including downsampling [20], blurring [103], and pixelation [41]. Later, encryption in alternate domains such as JPEG bitstreams [82, 101] and DCT coefficients [114, 115] was explored, though these often introduced severe distortions that hindered downstream tasks. Homomorphic encryption [110, 124] addresses inference on encrypted images, but suffers from high computational cost [78] and limited compatibility with advanced models like Vision Transformers [24, 93, 95]. Additionally, approaches for face images [12, 34, 69] leverage facial priors from StyleGAN [53] or face recognition networks [22, 118], which may not readily generalize to other domains, such as the medical-domain images in our work.

Medical Image De-Identification. Early methods (e.g., FreeSurfer [30], PyDeface [32], SynthStrip [44]) focus on removing facial features in brain MRI. For common medical images, early approaches use pixel-domain filters (like blurring [103] and pixelation [41]) or frequency-domain techniques [31], but these hand-crafted solutions also severely degrade the image details, leading to substantially degraded performance and visual quality [59, 60, 62]. Differential Privacy methods [19, 27, 57, 112] inject noise into the training data, which compromises inference-time utility. More recent generative models [15, 26, 43, 91, 109] synthesize images conditioned on disease labels or lesion masks. However, they tend to lack task generalizability and struggle to balance privacy-utility trade-offs, which are addressed by our approach.

Feature Decoupling. Variational auto-encoder (VAE)-based works [40, 88] decouple representations, by constraining the variables in latent space independent. Generative adversarial network (GAN)-based methods [14, 63] are unsupervised, leaving factors unaligned with explicit semantic or identity information. For face images, there are methods [23, 53, 54, 100] targeting identity separation. However, these methods rely on strong facial priors that may not generalize to medical images. In contrast, motivated by the explicit entropy regularization terms adopted in data compression field [2, 45–47, 61, 92, 94, 96–98], our approach effectively decouples identity in medical images, within a minimum-codelength space.

3. Methodology

In this section, we first describe the medical re-identification (ReID) models used for privacy attacks. Next, we introduce our de-identification model, which divides the task into two stages. First, identity information is removed via region blocking with an adjustable threshold. Second, lost medical semantics are compensated. This approach flexibly adjusts privacy while preserving rich, generalizable medical features for downstream tasks.

3.1. Medical ReID Models

Given a query medical image, ReID models aim to retrieve all images belonging to the same individual, from a medical record database. Concretely, the model first extracts identity

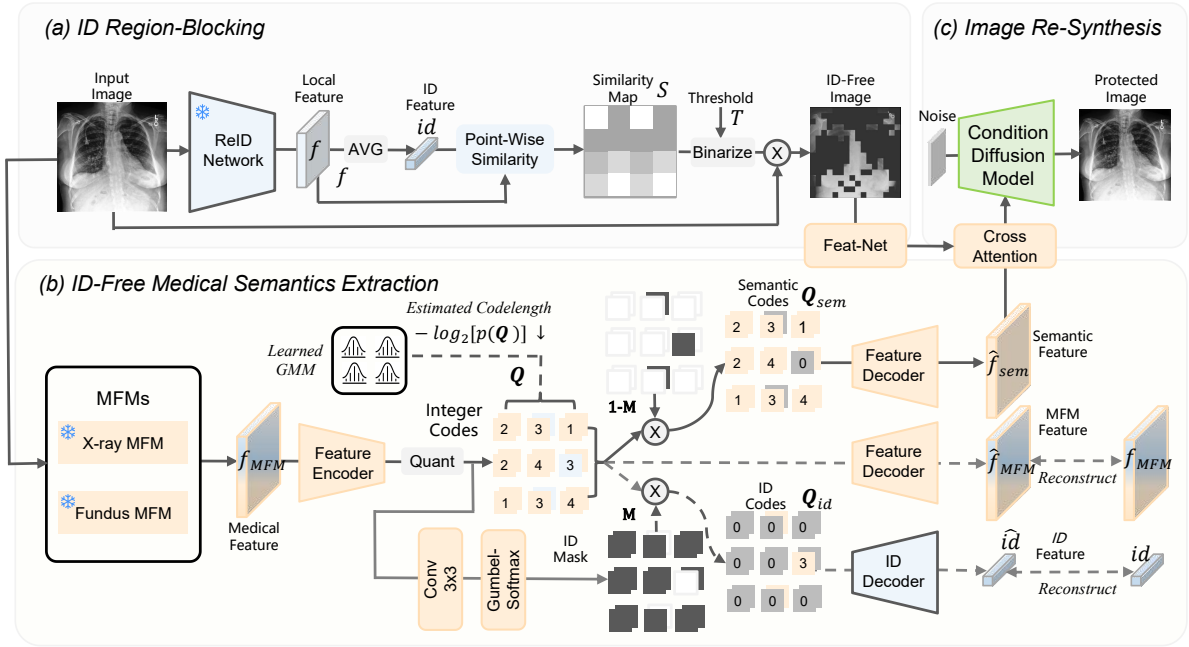

Figure 2. Overview of the proposed divide-and-conquer framework, DCM-DeID. (a) *ID-Blocking*: A pre-trained ReID network produces the identity-similarity map, which is binarized by different thresholds to adjust privacy level. (b) *ID-Free Medical Semantics Extraction*: Medical foundation models (MFMs) extract features that are encoded into a code space under the minimum-codelength regularization. A learned mask partitions the codes into identity- and medical semantics-related ones, where only the latter one is preserved. (c) *Image Re-Synthesis*: A diffusion model re-synthesizes images that are privacy-preserving and semantics-rich, generalizing to various downstream tasks. We illustrate with X-ray images, but the framework is also applicable to other modalities such as fundus images. ❄ denotes frozen models, while gray dashed lines indicate components used solely for learning identity-semantic decoupling. The channel number of codes Q is arbitrary; two channels are shown for conciseness. $\otimes$ denotes the element-wise multiplication.

(ID) embedding from the query image, and then compare it with that of each image within the database. Then, the image with the closest Euclidean distance is adopted as the re-identified image. We build two medical ReID models, i.e., ViT [24] and VisionMamba [122]-based ones, which are separately adopted in the training and the evaluation stages. These ReID models are optimized with a combination of classification loss and triplet loss [39], following the previous object ReID work [38].

3.2. A Divide-and-Conquer Approach

To defend against attacks on medical ReID models, we propose DCM-DeID, a divide-and-conquer approach for medical image de-identification. DCM-DeID operates in three stages: *ID Blocking*, which removes identity-related image regions; *ID-Free Medical Semantics Extraction*, which extracts rich medical information without reintroducing identity information; and *Image Re-Synthesis*, which generates the final de-identified medical image.

ID-Blocking. Given an input image $X \in \mathbb{R}^{3\times H\times W}$, where H and W denote the image spatial scales, we use a ViT-based ReID model to extract local features $f \in \mathbb{R}^{768\times h\times w}$, where $h = H/16$ and $w = W/16$. Spatial average pooling is applied to f to obtain an identity embedding $id \in \mathbb{R}^{768}$. For each spatial position in f, the cosine similarity with id is computed, resulting in a similarity map $S[i] = \cos(f[i], id)$, where i denotes the spatial location. Then, S is binarized by a threshold T. Finally, the ID-blocked image is computed as: $X^{noID} = X \odot \mathrm{Upsample}(S > T)$, where Upsample denotes nearest-neighbor interpolation to match the resolution of S to X.

ID-Free Medical Semantics Extraction. Although X^{noID} effectively removes identity information, it inevitably distorts medical cues such as lung shadows. To amend this, we employ pre-trained medical foundation models (MFMs), e.g., MGCA [105] for X-ray images, to extract rich medical feature f_{MFM} from X. Since f_{MFM} contains both semantic cues and local details that may encode identity, we introduce a feature decoupling strategy (Section 3.3) to decouple and remove the identity information, yielding the identity-free semantic feature $\hat{f}_{sem}$.

Image Re-Synthesis. Given X^{noID} and $\hat{f}_{sem}$, a dual-conditioned diffusion model synthesizes the de-identified image that inherits the rich semantics within MFMs, while also protecting privacy. Since the synthesized image is highly realistic, it can be directly deployed to downstream medical AI applications, without further adaptation. The model details are elaborated in Section 3.4.

3.3. Medical Semantics Decoupling

Medical features extracted by the MFM encode both diagnostic semantics (e.g., lesion morphology) and identity-

related cues (e.g., rib patterns in chest X-rays). For effective privacy-preserving, it is imperative to decouple these two types of information, and discard the identity cues. We achieve this by learning a minimum-length code space, and separating the two parts in this space.

Theoretical Motivation. From an information-theoretic perspective, the Minimum Description Length (MDL) principle [4, 33] states that the best representation for a given set of data is the one that minimizes the total codelength needed to describe the data, where each group of features tends to capture the independent or low-correlation information parts. In our context, let $\boldsymbol{Q}$ be the latent representation of the MFM feature f_{MFM} and let $H(\boldsymbol{Q})$ denote its expected codelength. The MDL principle objective can be seen as balancing a reconstruction loss and a compression term, i.e., the so-called rate-distortion loss (RD loss) [5]:

$$\mathcal{L}_{\text{code-all}} = \min_{\mathcal{E},\mathcal{D}} \quad \underbrace{\|f_{MFM} - \hat{f}_{MFM}\|_2}_{\text{Feature Reconstruction}} + \underbrace{\beta H(\boldsymbol{Q})}_{\text{Codelength}}, \tag{1}$$

where $\boldsymbol{Q} = \mathcal{E}(f_{MFM})$, $\hat{f}_{MFM} = \mathcal{D}(\boldsymbol{Q})$, and β denotes balancing weight. $\mathcal{E}$ and $\mathcal{D}$ represent a pair of feature encoder and decoder networks.

Discrete Code-based Codelength Estimation. Directly calculating the $H(\boldsymbol{Q})$ for the continuous variable $\boldsymbol{Q}$ is non-trivial [68]. Fortunately, the neural data compression community [2, 3, 70, 72] have verified that the codelength of *integer* latent variables can be quite precisely estimated with a *learnable entropy model*. Therefore, we append the quantization operation at the tail of the encoder $\mathcal{E}$, to make elements within $\boldsymbol{Q}$ discrete values, and estimate its codelength.

Concretely, $\mathcal{E}$ comprises three residual blocks [36] with 256 channels, followed by a convolutional layer to reduce dimensionality and a rounding operation that outputs a 32-channel integer code $\boldsymbol{Q}$. The decoder network $\mathcal{D}$ is symmetric to $\mathcal{E}$, except it omits the rounding operation. During training, the straight-through estimator [71] is employed to backpropagate gradients through the rounding step.

Following [2], the expected codelength of encoding $\boldsymbol{Q}$ is calculated as the log-likelihood, i.e., $H(\boldsymbol{Q}) = -\log_2 p(\boldsymbol{Q})$, where the probability $p(\boldsymbol{Q})$ is modeled using a Gaussian Mixture Model (GMM) [84] with K components:

$$p(\boldsymbol{Q}) = \sum_{k=1}^{K} w^k \cdot \mathcal{N}\left(\boldsymbol{Q}; \mu^k, e^{\sigma^k}\right), \tag{2}$$

where $\{\mathbf{w}, \boldsymbol{\mu}, \boldsymbol{\sigma}\}$ are the learnable mixture weights, means, and log variance scalers of the GMM components, respectively, which are shared across spatial positions, not unshared along the channel axis [2]. Following [16], K is set to three. For each integer element $q \in \boldsymbol{Q}$, the probability is computed over the quantization bin [18, 72], $p(q) = \mathcal{F}(q + 0.5) - \mathcal{F}(q - 0.5)$, where $\mathcal{F}(x) = \sum_{k=1}^{K} w^k \, \Phi\left(x; \mu^k, e^{\sigma^k}\right)$ is the cumulative distribution function (CDF) of the Gaussian Mixture Model (GMM), $\Phi\left(x; \mu, e^{\sigma}\right) = \frac{1}{2}\left[1 + \operatorname{erf}\left(\frac{x-\mu}{\sqrt{2}\, e^{\sigma}}\right)\right]$. We not that the CDF can be efficiently calculated by the modern deep learning framework such as PyTorch [80].

Learning of Identity-Associated Code Mask. A single convolution layer predicts a binary mask $\mathbf{M}$ from $\boldsymbol{Q}$, with the same dimensions as $\boldsymbol{Q}$. The Gumbel-Softmax algorithm [50] is applied to enable gradient propagation through the binary mask. The identity-associated codes are then obtained by element-wise masking, $\boldsymbol{Q}_{\text{id}} = \boldsymbol{Q} \odot \mathbf{M}$. A lightweight convolutional network, composed of three residual blocks followed by average pooling, predicts the identity embedding $\hat{id}$ from $\boldsymbol{Q}_{\text{id}}$. Then, the RD loss for reconstructing identity can be given by:

$$\mathcal{L}_{\text{code-id}} = \|\hat{id} - id\|_2 + \beta H(\boldsymbol{Q}_{\text{id}}), \tag{3}$$

where $H(\tilde{\boldsymbol{Q}}_{\text{id}})$ is calculated similarly to $H(\boldsymbol{Q})$, sharing the same GMM parameters and balancing weight β as in Equation 1, since they operate in the same latent space.

Reconstruction of Medical Semantics. By suppressing identity-related codes via the inverse mask $(1-\mathbf{M})$, we obtain the semantics-part codes $\boldsymbol{Q}_{sem} = (1-\mathbf{M}) \otimes \boldsymbol{Q}$. Finally, the final ID-free medical semantic feature is reconstructed as: $\hat{f}_{sem} = \mathcal{D}(\boldsymbol{Q}_{sem})$, which preserves critical diagnostic semantics, excluding the identity information.

3.4. Image Re-Synthesis Model

Given the ID-masked image X^{noID} and the ID-free medical semantic feature $\hat{f}_{sem}$, we employ a diffusion model to synthesize de-identified medical images. First, we utilize a Feat-Net to project the high-resolution X^{noID} into the low-resolution feature $f^{noID} \in \mathbb{R}^{512 \times \frac{H}{32} \times \frac{W}{32}}$. The Feat-Net consists of the VAE encoder from Stable Diffusion [86], followed by two convolution layers of kernel size 5 and stride size 2. Next, we adopt a bi-directional cross-attention mechanism [13] to fuse f^{noID} and $\hat{f}_{sem}$, producing a fused feature $f_{fuse} \in \mathbb{R}^{512 \times \frac{H}{32} \times \frac{W}{32}}$, which is further processed through a series of convolutional layers. This produces a set of features with dimensions matching those of the UNet's intermediate feature maps within the diffusion model. These features are added to the UNet layers, guiding the diffusion process toward two objectives: maintaining the privacy level of X^{noID}, while preserving the medical semantics in $\hat{f}_{sem}$.

3.5. Learning Strategy

The whole framework is end-to-end optimized, with the following objective,

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{code-all}} + \mathcal{L}_{\text{code-id}} + \mathcal{L}_{\text{diffuse}}, \tag{4}$$

where $\mathcal{L}_{\text{diffuse}}$ denotes the diffusion loss [42]. We do not introduce the balancing weight, since we found directly adding the loss terms already achieves satisfactory results.

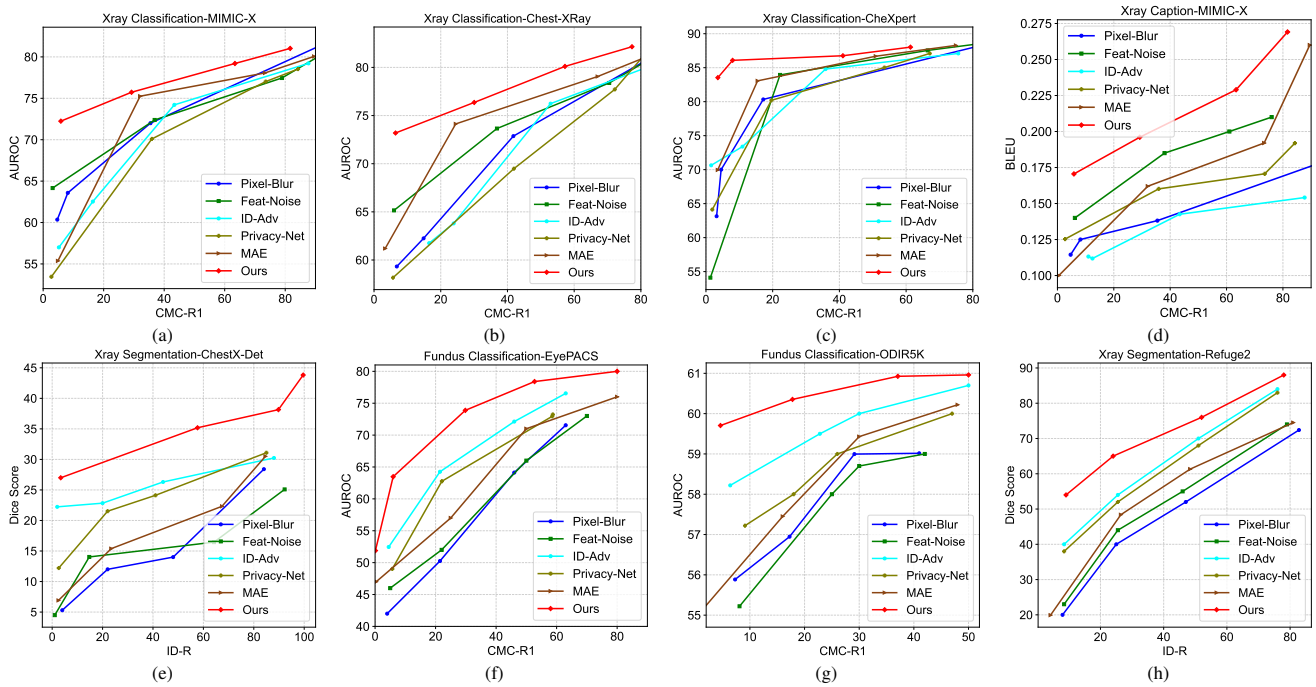

Figure 3. Identity-performance trade-off curves of various medical privacy protection methods.

Attack SR	Method	X-ray Classify AUROC (%)			X-ray Caption BLEU	X-ray Seg Dice (%)	Fundus Classify AUROC (%)		Fundus Seg Dice (%)
		MIMIC-X	Chest-XRay	CheXpert	MIMIC-X	ChestX-Det	EyePACS	ODIR5K	Refuge2
10%	Pixel-Blur [103]	64.15	60.48	74.45	0.1259	7.53	44.83	56.17	22.35
	Feat-Noise [116]	65.84	66.25	66.49	0.1453	10.74	47.76	55.53	24.85
	ID-Adv [77]	59.36	61.75	73.11	0.1131	22.23	56.28	58.50	41.23
	Privacy-Net [55]	57.09	59.51	71.47	0.1329	15.75	52.63	57.29	39.23
	MAE [37]	59.11	65.26	77.05	0.1186	9.94	50.91	56.50	27.33
	Ours	**72.86**	**73.66**	**86.12**	**0.1750**	**27.98**	**65.23**	**59.96**	**54.66**
20%	Pixel-Blur [103]	67.23	64.27	80.67	0.1307	11.25	49.56	57.42	34.11
	Feat-Noise [116]	68.27	69.01	80.78	0.1595	14.28	51.29	57.17	37.20
	ID-Adv [77]	64.08	62.71	77.53	0.1193	22.83	63.24	59.28	49.47
	Privacy-Net [55]	62.11	62.63	80.25	0.1434	20.55	61.01	58.25	47.47
	MAE [37]	66.43	71.38	83.52	0.1383	13.99	54.97	58.01	40.13
	Ours	**74.35**	**75.01**	**86.32**	**0.1859**	**29.49**	**69.59**	**60.41**	**62.04**
	Original	82.13	84.82	87.24	0.3218	52.89	81.46	61.53	90.08

Table 1. Performance comparison of medical image privacy protection methods, under different attack Success-Rates (SR). For measuring SR, we adopt CMC-R1 metric for MIMIC-X, Chest-Xray, CheXpert, EyePACS, and ODIR5K, using ID-R metric for ChestX-Det and Refuge2. Original denotes the performance on original images, which is the performance upper-bound of privacy-removal images.

4. Experiments

4.1. Implementation Details

For *Med-ReID* models, we adopt the AdamW optimizer [66] during training, with a learning rate of 1e-5 scheduled by cosine decaying strategy and a weight decay of 1e-2. The training process consists of 300,000 steps. The batch size is 256. We apply random cropping and blurring as image augmentation strategies, and the input image resolution to networks is 256×256. The ViT-based models are initialized with CLIP-pretrained weights [83], while the VisionMamba-based models are initialized with ImageNet-pretrained weights [21]. Training a single ReID model takes about 24 hours with four NVIDIA RTX 4090 GPUs.

For *DCM-DeID* model, the UNet within the diffusion model follows the same architecture as the Stable Diffusion [86], also performing the diffusion procedure in the latent space. The feature channels within UNet are reduced to [128, 256, 512, 1024], for the four stages of both the down-pathway and up-pathway, to reduce computational cost. The identity-similarity map threshold T is defined as the r-th quantile of the similarity map S. r is selected from [0.95, 0.7, 0.4, 0.2] to cover wide privacy levels. We adopt MGCA-ResNet [105] and RetFound-ViT [121] MFMs for X-ray and fundus images, respectively. During training, we apply random flipping and random cropping 256×256 patches for data augmentation. The codelength loss term weight β is set to 0.5. At test time, we resize the shorter

side of the images to 256 and then center-crop the middle 256×256 region. The learning rate is set to 1e-4 and is gradually decayed with the cosine annealing strategy [67]. The total number of training steps is 800,000. The mini-batch size is 64. We utilize the AdamW optimizer [66] implemented in PyTorch [80] with CUDA support. The momentum parameters are set as $\beta_1 = 0.9$ and $\beta_2 = 0.99$, and the gradient norm is clipped to a maximum value of 1. The entire training process takes about three days on a machine equipped with eight NVIDIA RTX 4090 GPUs.

4.2. Datasets

We evaluate our approach on two medical image modalities: chest X-rays and eye fundus photographs, with seven public datasets. For the *chest X-ray* modality, we split the MIMIC-X dataset [51] into training, validation, and test sets using an 8:1:1 ratio. For the Chest-Xray and CheXpert datasets, we randomly select 10% patients as the test set. We also adopt the ChestX-Det dataset [64] to evaluate the X-ray segmentation task. For the *eye fundus* modality, we divide the EyePACS dataset [25] into training, validation, and test sets with an 8:1:1 ratio, and we use the Refuge2 dataset [29] to evaluate the fundus segmentation task. ODIR5K [8] is also adopted for evaluating the fundus classification task of systemic diseases such as hypertension. Note that only MIMIC-X and EyePACS are used during training; all other datasets, which differ in environment, demographics, and imaging devices, are never seen during training, to evaluate the domain generalizability of our approach. MIMIC-X, CheXpert, Chest-Xray, EyePACS, ChestXDet, and ODIR5K contain 377K, 224K, 112K, 88K, 3.6K, and 5K images.

4.3. Reproduced Privacy Protection Methods

We implement several privacy protection methods, comparing them with our approach in a fair setting.

Pixel-Blur [103]. This method applies a Gaussian blur to the input image. We experiment with standard deviations of $\{1, 5, 10, 20\}$ to vary the level of de-identification.

Feat-Noise [116]. We train an autoencoder [116] and inject Gaussian noise into its latent features. The noise level is selected from $\{0.1, 0.8, 0.85, 0.9, 0.95\}$.

ID-Adv [77]. A UNet is trained to generate a de-identified image Y from the original image X, optimizing the loss $\mathcal{L} = \lambda \cos(id_X, id_Y) + \|med_X - med_Y\|_2 + \mathcal{L}_{reg}$, where id_X and id_Y are identity features extracted by a ViT-based ReID model, and med_X and med_Y are medical features obtained from MFMs same as our approach. $\mathcal{L}_{reg}$ is a GAN regularization loss ensuring visual plausibility, $\cos(\cdot, \cdot)$ denotes cosine similarity, and $\|\cdot\|_2$ the ℓ_2 norm. The trade-off weight λ is chosen from $\{0.1, 0.5, 1, 2\}$.

Privacy-Net [55]. This method updates the identity model and the de-identification network adversarially, enhancing de-identification performance. The original Privacy-Net focuses solely on segmentation tasks, supervised by segmentation masks. To enable task-agnostic de-identification, we train it using the same objective as ID-Adv. Since the identity model is adversarially updated and are stronger, we use smaller λ values compared to ID-Adv, i.e., $\{0.05, 0.25, 0.5, 1\}$.

MAE [37]. Following [108], we transfer the concept of masked auto-encoders (MAE) [37] to the adjustable privacy protection problem, by masking a random proportion of patches to obscure identity information. It adopts the same diffusion model as our approach to generate the masked regions. This model can also serve as a degenerated version of our model, where both semantic compensation and identity-region similarity designs are removed.

4.4. Downstream Task Models

For the *identity recognition*, we adopt the VisionMamba-based ReID model, which differs from the ViT-based model employed during the training of privacy protection methods, ensuring the method generalization capability across different ReID models. For the *X-ray classification*, we use the ViT model pre-trained with Med-UniC [104]. For *X-ray captioning*, we employ the visual-language model CXR-LLaVA-v2 [58], which is specifically designed for X-ray images. For *X-ray segmentation*, we adopt CGRSeg [75]. For *fundus classification*, we use the ViT model pre-trained with KeepFit [111]. Finally, for *fundus segmentation*, given the limited dataset size, we employ nnUNet [49].

4.5. Evaluation Metrics

For privacy evaluation, we adopt the cumulative matching characteristics (CMC) [10] at Rank-1, i.e., CMC-R1, on datasets with patient ID information available (i.e., MIMIC-X, Chest-Xray, CheXpert, EyePacs, and ODIR5K). For datasets without patient ID information (i.e., CheX-det and REFUGE2), we adopt the recognition rate, i.e., ID-R, which determines whether the distance between the ID feature of the original and de-identified image exceeds a predefined threshold. The thresholds are set to 1.1 and 1.35 for the X-ray and fundus modalities, respectively, based on statistics from the validation sets of MIMIC-X and EyePACS. For the disease diagnosis task, we employ the area under the receiver operating characteristic curve (AUROC) metric [11]; for the image captioning task, we use the bilingual evaluation understudy (BLEU) metric [79]; and for the image segmentation task, we adopt the Dice score metric [7].

4.6. Results

X-ray Classification. As shown in Table 1, our method substantially outperforms other approaches, achieving AUROCs of 72.86%, 73.66%, and 86.12% on MIMIC-X, Chest-XRay, and Chexpert, respectively, under CMC-

R1=10%. Notably, although our model is trained on MIMIC-X, it generalizes well to the other two datasets.

Among the other compared approaches, Feat-Noise obtains the second-best performance, i.e., AUROC of 65.84% at CMC-R1=10% on MIMIX-X, by condensing image pixels into a compact latent feature space. In contrast, methods that jointly optimize a trade-off between de-identification and medical preservation, i.e., ID-Adv and PrivacyNet, yield unsatisfactory performances. As shown in Figure 3 (a), under the ID-R=5% setting, ID-Adv and PrivacyNet attain AUROCs of only 56.32% and 54.21% on MIMIC-X, respectively, which are much lower than the simple pixel blurring baseline (61.62%). This indicates that directly optimizing the two conflicting objectives is suboptimal. In contrast, our approach decouples the objectives into two separate steps, identity removal and medical semantic compensation, achieving consistently superior performance.

As for MAE, which employs the same diffusion model as ours, it achieves competitive results at a high attacking rate, with an AUROC of 76.12% @CMC-R1=40% on MIMIC-X, outperforming all other approaches except ours. However, at a low attacking rate CMC-R1=10%, it falls behind our method by over 13% AUROC. This highlights that our superior performance is not solely due to the generative power of the diffusion model, but rather stems from the effectiveness of our core idea of semantic compensation.

X-ray Caption. As shown in Table 1, our method attains a BLEU score of 0.1750, remarkably surpassing Pixel-Blur (0.1259), Feat-Noise (0.1453), and MAE (0.1186), at CMC-R1=10%. This proves that our approach can comprehensively preserve the clinic-required information, beyond only the classification label.

X-ray Segmentation. Furthermore, we evaluate the methods on a fine-grained task: segmentation. As shown in Table 1, at ID-R1=10%, our method achieves a Dice score of 27.98%, outperforming Pixel-Blur (7.53%), Feat-Noise (10.74%), Privacy-Net (15.75%), ID-Adv (22.23%), and MAE(9.94%). This proves that our semantic compensation scheme not only preserves the global semantics for classification, but also effectively retains the local semantics for segmentation. Pixel-Blur and Feat-Noise perform poorly, since they severely corrupts the image details. In contrast, ID-Adv and Privacy-Net, which incorporate a medical feature-matching loss, achieve slightly decent performance, but still lag far behind our approach. For instance, under ID-R=80%, our method outperforms Privacy-Net by approximately 8%, as shown in Figure 3(e).

Fundus Classification. Beyond X-ray images, our method also proves effective on fundus data. For instance, on EyePACS and ORID5K, our approach outperforms the second-best competitor ID-Adv by about 9% and 1%, respectively. These results confirm that our approach generalizes well across different imaging modalities.

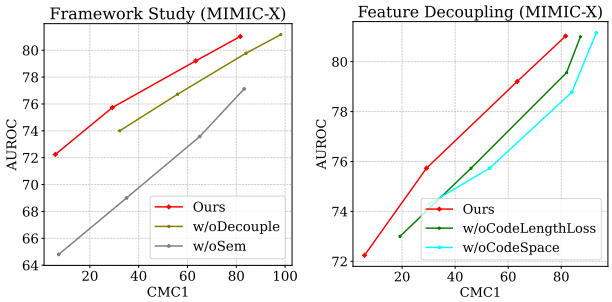

Figure 4. (*Left*) Ablation on the framework design. (*Right*) Ablation study on the feature decoupling strategy.

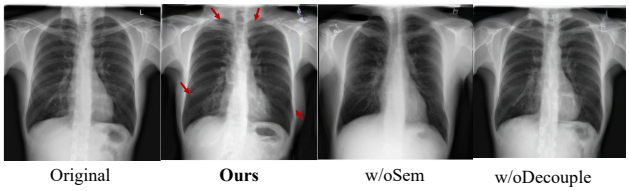

Figure 5. Qualitative comparison of different variant models. The models are described in Figure 4 caption. Red arrow denotes the modified identity-related features. Best to view by zooming-in.

Fundus Segmentation. Our method achieves a Dice score of 54.66% on REFUGE2 at ID-R1=10% , largely surpassing MAE (27.33%), Privacy-Net (39.23%), and ID-Adv (41.23%). This further validates that our method also effectively preserves fine-grained semantic cues of eye fundus.

4.7. Model Analysis

Framework-Level Ablation Study. As shown in Figure 4 (*Left*), by removing the semantic branch, the AUROC of the resulted model 'w/oSem' dramatically drops by over 8%, at CMC-R1=5%. On the other hand, without the identity-semantics decoupling mechanism, the resulting model 'w/oDecouple' leads to about 15% CMC-R1 increase, for achieving the similar AUROC performance, since substantial identity cues are leaked from the vanilla medical features of MFMs. We further illustrate the protected images from different models. As shown in Figure 5, our results effectively modify identity-related features, such as the shape and location of the clavicle and chest contour. The 'w/oSem' model also removes these regions but significantly alters medical manifestations. In contrast, the 'w/oDecouple' model preserves medical features but fails to sufficiently suppress identity-related features, such as clavicle shape, due to residual identity information in the features from MFMs. These results confirm that both medical semantics and identity-semantics decoupling are essential for our advanced medical DeID approach.

Besides, we attempt to remove ID-Blocking module, tuning privacy via activating more decoupled ID codes. This leads to degraded AUROC (72.01%,73.47%,74.68%), compared to using ID-Blocking (72.68%,74.35%,79.62%) on MIMIC-X (10%,20%,40% at tack SR settings), due to

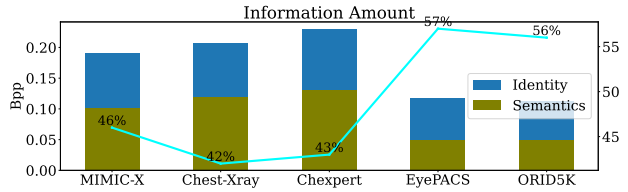

Figure 6. Comparison of semantic and identity information in terms of Bits-per-Pixel (bpp) [65], calculated as the feature codelength divided by the original image size.

the complete loss of pixel-wise details.

Ablation Study on the Decoupling Strategy. As shown in Figure 4 (*Right*), omitting the codelength loss terms ('w/oCodeLengthLoss') fails to effectively remove identity information from MFM features, since the loose space cannot effectively decouple the identity and the semantics information. Moreover, removing the discrete code bottleneck ('w/oCodeSpace') further exacerbates identity leakage, leading to further inferior performance.

Furthermore, we quantitatively compare the overall and identity-related information in MFM features, as shown in Figure 6. First, we notice that a significant portion is identity-related, i.e., around 44% and 55% for X-ray and fundus images. Second, the average information amount of the X-ray dataset Chest-Xray is 0.23bpp, much higher than 0.11bpp achieved by the fundus dataset EyePACS. This is aligned with the medical knowledge prior, that X-rays capture multiple organs and tissues, containing much complex information, than the fundus image that only focuses on eyes. This proves that the learned codelength effectively describes the medical data characteristics.

Finally, we analyze the impact of the codelength loss weight β and the latent code channel number. As shown in Figure 7 (*Left*), reducing β from 0.5 to 0.1 significantly increases CMC-R1 from 5.85% to 12.34%, as a loosely constrained code space fails to effectively decouple identity information. Conversely, increasing β from 0.5 to 2 has little effect on CMC-R1 but reduces AUROC performance by approximately 6%, as an overly strong constraint impairs semantic feature reconstruction. The number of code channels also influences performance, by tuning the information capacity of the latent code, as shown in Figure 7 (*Right*). However, since β directly regulates the code-length term, the impact of the channel number is limited.

Discussion with Label-Conditioned Diffusion Models. These methods [48, 107, 120] employ task-specific labels (e.g., disease labels or text reports) to synthesize images, which are limited to label-associated tasks. In contrast, our approach is task-agnostic and applicable to diverse tasks. Moreover, after fine-tuning our approach towards a single task, i.e., replacing the MFM with a supervised classification network, our method achieves 81.92% AUROC at CMC-R1 = 0.30%, surpassing the label-conditioned model,

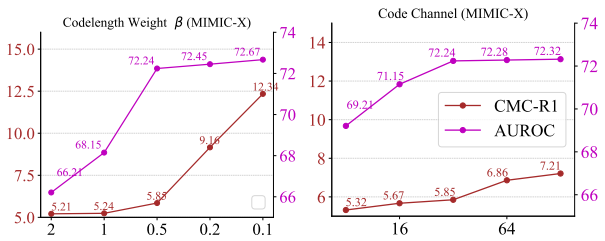

Figure 7. (*Left*) Impact of the rate-distortion weight β. (*Right*:) Impact of the code dimension. All experiments are evaluated by masking 95% identity-related regions, for a fair comparison.

i.e., 80.79% AUROC at CMC-R1 = 0.29%. This confirms that our minimum-codelength representation also benefits the single-task setting, compared to the methods directly using the task labels guiding the diffusion procedure.

Model Complexity. All methods and our model comprise about 380M parameters, for a fair comparison. Our inference time is 540 ms on an NVIDIA 4090 GPU, which is similar to MAE (526ms), but slower than Privacy-Net (120 ms), ID-Adv (122 ms), and Feat-Noise (124 ms), due to the multiple inference steps of diffusion procedure. Nonetheless, given the significant performance gains and that the medical imaging procedure itself is time-consuming, the running time is acceptable and does not hinder clinical workflows. In the future, we will integrate the single-step diffusion technique [113] to accelerate the process.

5. Conclusion, Future Works, and Other

Conclusion. We have presented DCM-DeID, a divide-and-conquer framework for medical image de-identification. By leveraging pre-trained Medical Foundation Models and a minimum codelength-based feature decoupling strategy, our method effectively remove identity cues, while preserving medical task utility. Extensive evaluations demonstrate the superiority of our approach. **Future Works.** We will extend our approach to multi-slice images, such as those produced by Magnetic Resonance Imaging (MRI). **Broader Impacts.** Our DeID technique is designed for medical AI applications, aiding the human. We emphasize that all rigorous clinical decisions must be made by human physicians using the original medical images. Furthermore, it is critical to enforce ethical guidelines, working in synergy with technological approaches to achieve medical privacy protection.

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